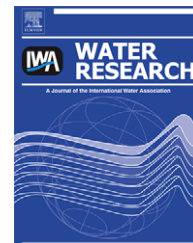


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Photoelectrocatalytic COD determination method using highly ordered TiO₂ nanotube array

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ABSTRACT

This work focuses on the experimental studies of a photoelectrocatalytic method for COD determination in a thin-cell reactor based on a highly effective TiO₂ nanotube array electrode. The effect of preparation parameters on the photoelectrocatalytic performance of TiO₂ nanotube array electrodes including the electrolyte, anodic potential, anodic time, solution pH, calcination temperature and time was examined. The TiO₂ nanotube array electrode prepared in preparation parameters at 1% HF electrolyte solution, anodic potential 20 V, anodic time 5 min, calcination temperature 450 °C with highly photoelectrocatalytic performance was chosen as the working electrode. When it is used in a thin-cell photoelectrocatalytic reactor for COD determination, it requires about 1–5 min to complete the oxidation of organics without further titration, much faster than the standard K₂Cr₂O₇ method (2–4 h). It consumes very limited harmless and inexpensive supporting electrolyte, free from secondary pollution. A wide dynamic working range of 0–850 mg/L can be achieved by this method, much wider than any other photoelectrocatalytic methods using TiO₂ nanoparticles or nanofilms in the reported literature. The effects of the water components were studied to propose the TiO₂ nanotube array method. Real sample analyses were also carried out and the COD value of real samples determined by this method agreed well with the standard dichromate method, and it shows good accuracy, stability and reproducibility.

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1. Introduction

The highly ordered TiO₂ nanotube arrays prepared by the anodization on Ti sheet have drawn a lot of attention over the past several years, largely because of their peculiar architecture, remarkable separation of photogenerated electron/hole pairs properties as well as the elevated photoelectrical and electrochemical properties. The TiO₂ nanotube arrays have wide-range applications (Varghese et al., 2006; Liu et al., 2007; Popat et al., 2007) in photocatalysis, photoelectrocatalysis, photovoltaic cell, degradation of organic pollutants, hydrogen sensors, and biocompatible materials. The nanotubular structure is vertical to the conductive Ti substrate, forming

a Schottky-type contact naturally and can provide an unidirectional electric channel for the transport of photogenerated electrons (Macak et al., 2005). For instance, the photo-conversion efficiency has come to 5.44% for dye-sensitized TiO₂ nanotube arrays solar cell (Premchand et al., 2006) and 16.25% for hydrogen generation by water photoelectrocatalytic reaction under UV irradiation (Lee et al., 2004). The degradation rate of pentachlorophenol on TiO₂ nanotube arrays electrode was found to be 1.86 times as high as TiO₂ film electrode (Quan et al., 2005).

With the water pollution problem getting more and more serious, water quality assessment and pollution control have become an important world concerned issue. Chemical

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oxygen demand (COD) is one of the most commonly used measurable parameters in assessing water quality and an important index for the control of operation of wastewater treatment plants (Fang and Chui, 1993; Brdjanovic et al., 2000; Liang et al., 2007). Though it is widely used, the standard method for COD determination, $K_2Cr_2O_7$ method has several drawbacks. It requires a time-consuming reflux process (2–4 h) for oxidation and consumes many expensive (e.g. Ag_2SO_4), corrosive (e.g. concentrated H_2SO_4), and toxic (e.g. Hg^{2+} , $Cr_2O_7^{2-}$) reagents (Hejzlar and Kopacek, 1990). In recent years, lots of efforts have been made to improve standard method, such as modified $K_2Cr_2O_7$ methods with microwave-assisted (Domini et al., 2006) or ultrasound-assisted oxidation (Domini et al., 2006), or to find a rapid, accurate and environmentally friendly method, such as electrocatalytic method with PbO_2 (Ai et al., 2004; Li et al., 2005) or Cu (Lee et al., 1999) electrode, photocatalytic (Kim et al., 2001; Chen et al., 2005; Zhu et al., 2006) and photoelectrocatalytic (Zhao et al., 2004) methods with TiO_2 materials. It is stipulated by RoHS directive of 2005/618/EC that all the electric machine or equipment containing Pb should be banned in EU market from 07/2006, so the practical application of PbO_2 method will face serious difficulty. Among these new developments, photocatalytic and photoelectrocatalytic methods are the most promising alternative due to the strong organics oxidation ability and green disposing processes of illuminated TiO_2 materials. Nevertheless, the photogenerated electron/hole pairs in random TiO_2 nanoparticles and nanofilm recombine easily. This drawback results in low photocatalytic activity which causes narrow dynamic working range and poor reproducibility in COD determination.

In this work, the photoelectrocatalytic activity of TiO_2 nanotube arrays prepared in different conditions was examined. The effect of preparation parameters on the photoelectrocatalytic performance of TiO_2 nanotube array electrodes including the electrolyte, anodic potential, anodic time, solution pH, calcination temperature and time was examined. A photoelectrocatalytic COD determination method was investigated by combining the optimized TiO_2 nanotube arrays, thin-cell reactor and photoelectrocatalytic techniques. The combination is favorable to facilitate the separation of photogenerated electron/hole pairs and enhance the photoelectrocatalytic degradation of various organics in the environment. The results show that this method overcomes many problems associated with the conventional COD determination methods such as long analysis time, consumption of expensive and toxic reagents, production of secondary toxic waste, and poor reproducibility.

2. Experimental

2.1. Materials

Ti sheet of 99.99% purity (0.1 mm thickness, 1.5 cm × 5 cm, Kurumi works, Japan). All chemicals in this study were of analytical grade. High-purity deionized water was used in the preparation of the solutions needed in the whole experiment.

2.2. Preparation of TiO_2 nanotube array electrode

The preparing method of TiO_2 nanotube array electrodes in this work was electrochemical anodization. Two different methods were chosen to prepare the TiO_2 nanotube array electrodes with different lengths. One method used an electrolyte containing 1% HF solution, with NaOH added to adjust the pH (Gong et al., 2001; Bai et al., 2008). The other used an electrolyte containing 0.1 mol/L KF, 1 mol/L $NaHSO_4$ and 0.2 mol/L trisodium citrate, also with NaOH added to adjust the pH (Cai et al., 2005). The TiO_2 nanotube array electrodes were prepared in a 1 L plastics vessel at room temperature, under constant stirring for different times. After anodization, the as-prepared sample was washed with high-purity deionized water and dried at room temperature. Then the as-prepared sample was annealed in a laboratory muffle furnace at 400, 450, 500, 600 °C to crystallize in the air ambience with the heating and cooling rate of 1 °C/min. The morphology of the annealed TiO_2 nanotube array was characterized by SEM using a microscope equipped with an energy-dispersive X-ray spectrometer (Philips, Netherlands, Sirion200). The crystal phase of the TiO_2 nanotube array was characterized with XRD (Bruker, Germany, AXS-8 Advance).

2.3. Determination of photoelectrocatalytic properties of TiO_2 nanotube array electrode

The determination of photoelectrocatalytic properties of TiO_2 nanotube array electrode was carried out in a rectangular quartz reactor (20 × 40 × 50 mm) with a platinum foil as a counter electrode, a saturated Ag/AgCl reference electrode and a TiO_2 nanotube array electrode as a working electrode. A 1 mol/L NaH_2PO_4 solution was chosen as the supporting electrolyte in the experiment. The working electrode is connected to the power supply with a clamp joint with a copper wire. The supply bias and work current are controlled by a CHI electrochemical analyzer (CHI 601b, CH Instruments, Inc., US). A 4 W UV lamp (GE, Japan, G4T5) was chosen as UV light source.

2.4. Apparatus and methods for photoelectrocatalytic COD determination

The TiO_2 nanotube array electrode annealed was used as the working anode for COD determination in the three electrode thin-cell reactor. The thin-cell reactor consisted of six main components (Fig. 1): a cell body, a TiO_2 nanotube array anode, a Pt counter electrode, a saturated AgCl/Ag reference electrode, a flow inlet, and a flow outlet. The cell body was assembled by compressing two teflon boards and a middle spacer (0.1 mm thickness) along with screw bolts and nuts. A quartz window (0.785 cm²) was embedded into the front teflon board and used to admit illumination. All interstices were sealed with silicon gel to keep the reactor airtight.

Photoelectrochemical measurements were performed using a CHI electrochemical analyzer. A 1 mol/L NaH_2PO_4 solution was chosen as the supporting electrolyte in the experiment. The concentration of 1 mol/L is able to provide adequate electrical conductivity to minimize the high

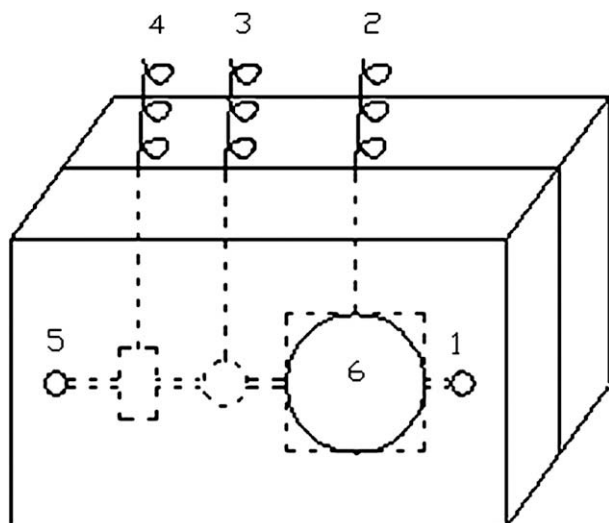


Fig. 1 – Schematic diagram of the thin-cell reactor: (1) flow inlet, (2) TiO_2 nanotube array electrode, (3) saturated AgCl/Ag reference electrode, (4) Pt counter electrode, (5) flow outlet, (6) quartz window.

resistance of narrow channel between the anodic and cathodic compartment in the thin-cell reactor.

3. Results and discussion

3.1. Characterization of TiO_2 nanotube array electrode

Fig. 2 shows the SEM images of TiO_2 nanotube array samples prepared in 1% HF electrolyte at pH 2 and a potential of 20 V for

3 different preparing times including 100 s, 300 s, 1200 s. The SEM images for nanotubes become highly ordered and well aligned when the preparing time is more than 300 s (5 min), the inner pore diameter is about 100 nm, and the length is approximately 350 nm for 5 min preparing time. Fig. 3 shows the SEM images of a TiO_2 nanotube array sample prepared in an electrolyte containing 0.1 mol/L KF, 1 mol/L NaHSO_4 and 0.2 mol/L trisodium citrate at pH 4 and a potential of 20 V for 8 h. The inner pore diameter is about 100 nm, and the length is approximately 1500 nm, and much longer than that prepared in an electrolyte containing 1% HF.

3.2. The photoelectrocatalytic properties of TiO_2 nanotube array electrode anodized in different electrolytes

The effect of preparation parameters including the anodic potential, anodic time, solution pH, calcination temperature and time on the saturated photocurrent density of TiO_2 nanotube array electrodes was obtained in continuous cyclic voltammograms, in the 1 mol/L NaH_2PO_4 electrolyte under the same UV irradiation. As shown in Table 1, the No. 1 electrode anodized in an electrolyte containing 1% HF has the highest saturated photocurrent density. As shown in Table 2, the No. 5 electrode anodized in an electrolyte containing 0.1 mol/L KF, 1 mol/L NaHSO_4 and 0.2 mol/L trisodium citrate has the highest saturated photocurrent density, but its photocurrent density is still lower than that of No. 1 sample anodized in an electrolyte containing 1% HF. Based on the above experimental results, the TiO_2 nanotube array electrode prepared in 1% HF electrolyte at pH 2, at the anodic potential of 20 V for 5 min, followed by the calcinations for 3 h at 450 °C (No.1 in Table 1) was chosen as the working electrode in the thin-cell reactor for COD determination.

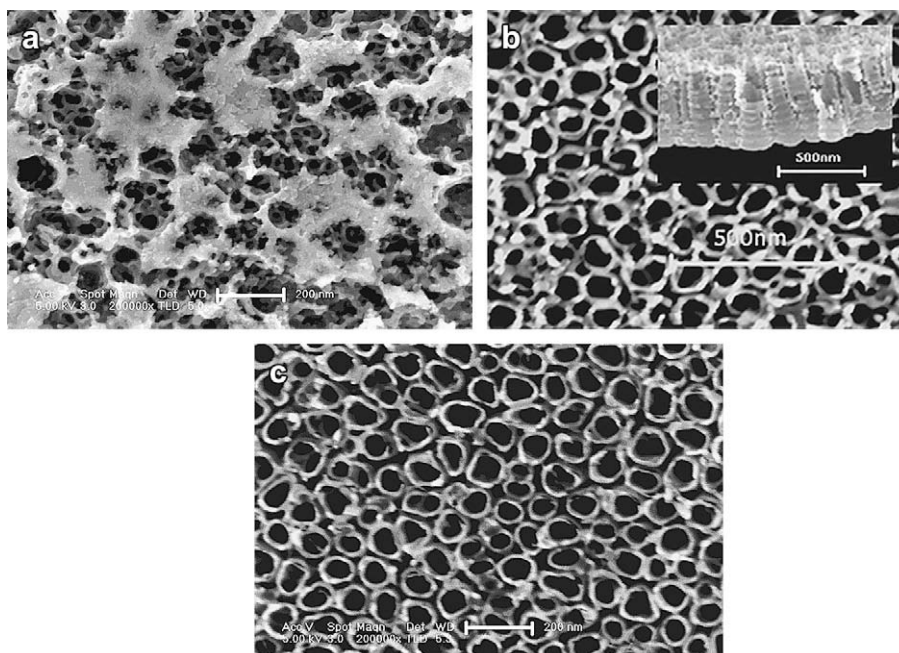


Fig. 2 – SEM images of a TiO_2 nanotube array sample prepared in electrolyte containing 1% HF at a potential of 20 V for different anodic times. a) 100 s, b) 300 s, c) 1200 s.

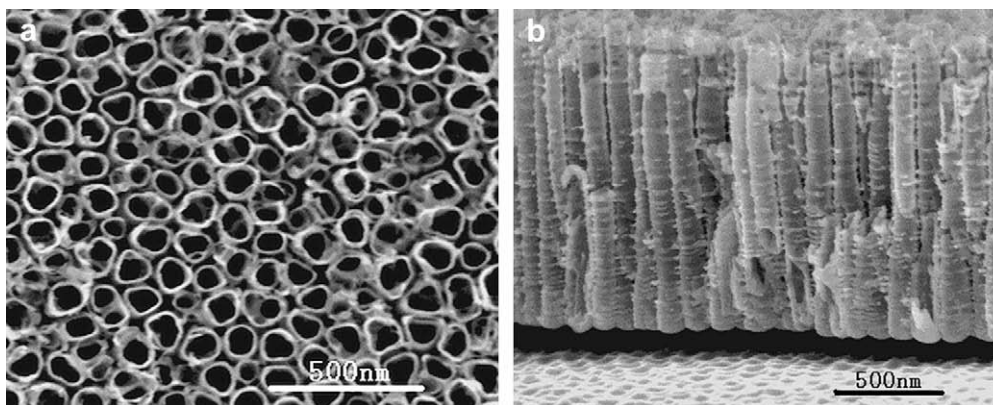


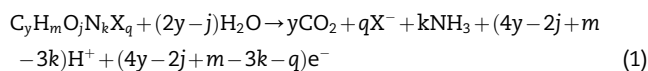
Fig. 3 – SEM images of a TiO₂ nanotube array sample prepared in electrolyte containing 0.1 mol/L KF, 1 mol/L NaHSO₄ and 0.2 mol/L trisodium citrate at a potential of 20 V for 8 h. a) Top view, b) cross-sectional view.

3.3. The stability and reproducibility of the TiO₂ nanotube array electrode

The accuracy and reproducibility of COD determination by photoelectrocatalytic method is dependent upon the stability of electrode. Continuous cyclic voltammograms (CV) was used to test the No. 1 TiO₂ nanotube array electrode in Table 1 under UV irradiation for five times with the scanning step of 0.02 V/s. The five CV curves are nearly superimposed upon each other, revealing that the photoelectrocatalytic activity of TiO₂ nanotube array electrode is very stable. Furthermore, the photocurrent of TiO₂ nanotube array electrode stabilizes at the bias potential of 0.5–3.0 V. This property is favorable to the determination of COD because within the range the photocurrent is less likely to be influenced by the fluctuation of the bias potential.

3.4. The analytical principle of COD determination using photoelectrocatalytic oxidation in the thin-cell reactor

The oxidation of organic compounds at the electrodes can be represented by the following electrode reaction equation:



where N and X represent a nitrogen and a halogen atom respectively. The numbers of carbon, hydrogen, oxygen, nitrogen and halogen atoms in the organic compound are represented by y, m, j, k and q.

Faraday's law can be used to quantify the concentration by measuring the charge passed if the organics is completely oxidized. That is

$$Q = \int I_{ph} dt = nFVC \quad (2)$$

where I_{ph} is the photocurrent from the oxidation of the organics and t is the time elapsed upon illumination. Q is the direct measure of net charge of the organics. n is the number of electrons transferred and C is the concentration of the organics, while the volume, V, and the Faraday constant, F are known constants.

Since one oxygen molecule is equivalent to 4 electrons transferred ($O_2 + 4H^+ + 4e^- \rightarrow 2H_2O$), the measured Q value can be easily converted to an equivalent O₂ concentration (or chemical oxygen demand). The equivalent of COD value can therefore be represented as:

$$COD \text{ (mg/L of } O_2) = \frac{nC}{4} \times 32,000 = \frac{Q}{4FV} \times 32,000 = KQ \quad (3)$$

Table 1 – The effect of preparation parameters on the saturated photocurrent density of TiO₂ nanotube array electrodes anodized in an electrolyte containing 1% HF, with sodium hydroxide added to adjust the pH.

Sample No.	pH	V (V)	t ₁ (min)	T (°C)	t ₂ (h)	Crystal type	PD (mA/cm ²)
1	2	20	5	450	3	A	0.66
2	2	20	20	450	3	A	0.58
3	2	20	40	450	3	A	0.51
4	3.5	20	5	450	3	A	0.55
5	3.5	20	20	450	3	A	0.60
6	3.5	20	40	450	3	A	0.57
7	2	20	5	500	3	A	0.65
8	2	20	5	600	3	A, R	0.35

A represents Anatase, R represents rutile. t₁, t₂ represent the anodic time and the calcination time respectively. V is the anodic potential and T is the calcination temperature. PD refers to the saturated photocurrent density of TiO₂ nanotube array electrode under the same irradiation of a 4 W UV lamp.

Table 2 – The effect of preparation parameters on the saturated photocurrent density of TiO₂ nanotube array electrodes anodized in an electrolyte containing 0.1 mol/L KF, 1 mol/L NaHSO₄ and 0.2 mol/L trisodium citrate, with sodium hydroxide added to adjust the pH.

Sample No.	pH	V (V)	t ₁ (h)	T (°C)	t ₂ (h)	Crystal type	PD (mA/cm ²)
1	4	20	4	400	3	A	0.409
2	4	20	8	400	3	A	0.423
3	4.5	20	8	400	3	A	0.415
4	4	20	8	500	3	A	0.446
5	4	20	8	500	6	A	0.470
6	4	20	8	600	6	A, R	0.345

It's evident from Eq. (3) that the concentration of organics is proportional to the charge transferred, so the COD value can be calculated by measuring Q . When different kinds of organics can be found in the sample:

$$\text{COD}_{\text{total}} = KQ_{\text{total}} \quad (4)$$

A set of typical photocurrent–time profiles obtained during an exhaustive degradation of organic compounds in the thin-cell reactor are shown in Fig. 4. The photocurrent ($i_{\text{background}}$) observed from the electrolyte solution is originated from water oxidation, while photocurrent (i_{total}) observed from the sample solution is the total current of two different components, one is originated from photoelectrocatalytic oxidation of organics and the other is originated from water oxidation, which is the same as the background photocurrent. When the i_{total} decreases to the level of $i_{\text{background}}$, the oxidation of organics is finished. For a given time period, $Q_{\text{background}}$ and Q_{total} can be obtained by integration of photocurrents with time. The net charge, Q_{net} , originated from the oxidation of organics, can be obtained by subtracting $Q_{\text{background}}$ from Q_{total} . Q_{net} then can be used to quantify the COD value of a sample solution. Experiment shows that the degradation rate of TiO_2 nanotube array anode upon organics is faster than the TiO_2 film anode that is comprised of nanoparticles. It requires about 1–5 min to complete the oxidation of organics without further titration, much faster than the standard $\text{K}_2\text{Cr}_2\text{O}_7$ method (2–4 h).

3.5. Validation of analytical principle

To validate the proposed method, a range of various organic compounds were chosen, including potassium hydrogen phthalate (KHP) and nicotinic acid. KHP is a standard reagent for testing the standard COD method, and nicotinic

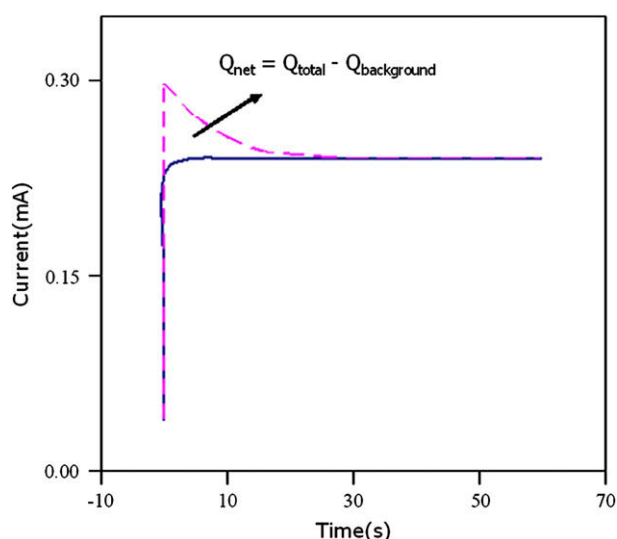


Fig. 4 – Photocurrent response of 1 mol/L NaH_2PO_4 blank solution (solid line) in the thin-cell reactor based on the TiO_2 nanotube array electrode and the photocurrent response of a 1 mol/L NaH_2PO_4 solution containing glucose (COD 140 mg/L) (dashed line).

acid is a kind of pyridine which is difficult to oxidize. Fig 5(a) is derived by plotting the net charge against theoretical COD values. The result shows that all organic compounds investigated are fitted into a linear line $y = (0.000187 \pm 0.0000027) x - (0.010240 \pm 0.00125)$ (where y represents the Q_{net} obtained from organics oxidation, x represents the COD value of the organics) with $R^2 = 0.9872$, which proves the proportional relationship between the net charge and COD value in an exhaustive degradation model. For different organics, both easy and difficult to oxidize, the net charges show good linearity with the theoretical COD values. This is the quantitative foundation for the photoelectrocatalytic method of COD determination. In Fig 5(b), the correlation between the net charge and the COD value for low concentrated samples suggests that this method can also be used to determinate the COD value of slightly polluted water (ground water or underground water).

3.6. Effect of the water components

Same as the study of other COD determination methods, the effects of the components of water sample were studied to propose the TiO_2 nanotube array method. In this work, three typical water components were investigated: sample solution pH, concentration of Cl^- (originated from NaCl) and concentration of NH_4^+ (originated from $\text{NH}_4\text{H}_2\text{PO}_4$). Glucose was chosen as the target organic, whose theoretical COD is 400 mg/L. The sample solution pH was adjusted with H_3PO_4 and NaOH .

Table 3 shows the effect of sample solution pH on the COD determination performance. It was found that within the sample solution pH range of 4–10, the COD value

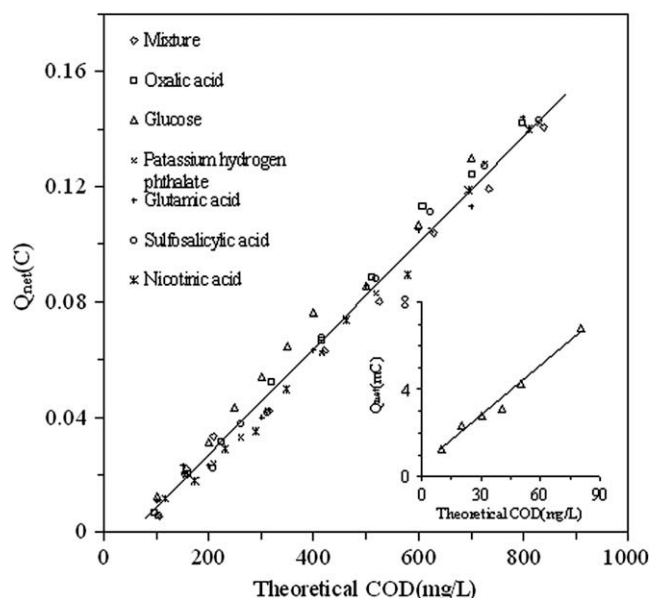


Fig. 5 – Validation of analytical principle: (a) correlation between the net charge and COD value; (b) correlation between net charge and COD value in low concentrated samples. The mixture is 6:2:1:1 sucrose and urea and tartaric acid and oxalic acid.

Table 3 – Effect of pH on the COD determination performance.

pH	Theoretical COD (mg/L)	TiO ₂ nanotube array method	
		COD (mg/L)	RD (%)
1	400	60.4	–84.9
2	400	255.0	–36.25
4	400	403.3	0.83
7	400	402.8	0.70
10	400	404.1	1.03
13	400	212.5	–46.88
14	400	45.6	–88.6

obtained using TiO₂ nanotube array method was essentially independent of pH variations and was nearly equal to the theoretical COD value. When the sample solution pH is below 4 or above 10, a sharp decrease in the COD value obtained using TiO₂ nanotube array method was observed and the TiO₂ nanotube array electrode was less stable when the sample solution pH was below 2 or above 13. These results suggest that the suitable pH range for the COD determination is 4–10.

Table 4 shows the effect of concentration of Cl[–] on the COD determination performance. It was found that within the concentration range of Cl[–] from 0 mg/L to 2000 mg/L, the COD value obtained using TiO₂ nanotube array method was nearly independent of C_{Cl[–]} variations and was nearly equal to the theoretical COD value. When the C_{Cl[–]} reached 2500 mg/L, the measurement of the COD value was irreproducible and the COD value obtained was significantly lower than the theoretical COD value. So if the C_{Cl[–]} of a sample solution reached 2500 mg/L, its Cl[–] needs to be removed by chemical method, such as using AgNO₃ to make Cl[–] become AgCl precipitate.

Table 5 shows the effect of concentration of NH₄⁺ on the COD determination performance. It was found that within the concentration range of NH₄⁺ from 0 mg/L to 2000 mg/L, the COD

Table 4 – Effect of concentration of Cl[–] on the COD determination performance.

C _{Cl[–]} (mg/L)	Theoretical COD (mg/L)	TiO ₂ nanotube array method	
		COD (mg/L)	RD (%)
0	400	402.2	0.55
50	400	403.7	0.93
100	400	397.3	–0.68
150	400	404.5	1.13
200	400	408.4	2.10
300	400	396.4	–0.90
400	400	406.0	1.50
500	400	411.8	2.95
700	400	413.4	3.35
1000	400	421.0	5.25
1500	400	428.7	7.18
2000	400	382.5	–4.38
2500	400	188.6	–52.85
3000	400	59.2	–85.20

Table 5 – Effect of concentration of NH₄⁺ on the COD determination performance.

C _{NH₄⁺} (mg/L)	Theoretical COD (mg/L)	TiO ₂ nanotube array method	
		COD (mg/L)	RD (%)
0	400	397.2	–0.70
50	400	402.5	0.63
100	400	404.0	1.00
150	400	404.5	1.13
200	400	398.9	–0.28
300	400	405.3	1.33
400	400	406.0	1.50
500	400	405.1	1.28
700	400	391.6	–2.10
1000	400	386.8	–3.30
1500	400	385.7	–3.58
2000	400	386.4	–3.40

value obtained was independent of C_{NH₄⁺} variations and was nearly equal to the theoretical COD value.

3.7. Photoelectrocatalytic COD determination of real wastewater

In this work, the standard COD determination method (APHA, 1998), K₂Cr₂O₇ method, was used to test the accuracy of the photoelectrocatalytic COD determination method. KHP was used to obtain the calibration graph.

The applicability of the photoelectrocatalytic COD determination method was further tested using eight different types of real wastewater. Table 6 shows that the test results from the TiO₂ nanotube array method show excellent correspondence with the K₂Cr₂O₇ method. The excellent correspondence provides compelling support for the suitability of the TiO₂ nanotube array method for the measurement of COD.

Table 6 – Comparison of COD values of various wastewater samples obtained by different methods.

Sample	K ₂ Cr ₂ O ₇ method	TiO ₂ nanotube array method	RD (%)
<i>Organic chemical plant wastewater</i>			
Sample 1	120.7	118.9	–1.49
Sample 2	118.7	111.4	–6.15
Sample 3	798.5	839.1	3.83
Sample 4	808.6	840.0	3.88
<i>Pesticide wastewater</i>			
Sample 5	573.4	547.4	–4.53
Sample 6	576.8	583.6	1.18
<i>Washing wastewater</i>			
Sample 7	19.8	20.5	3.54
<i>Petroleum exploiting wastewater</i>			
Sample 8	200	203.6	1.80
<i>Dyeing wastewater</i>			
Sample 9	43.1	41.3	–4.18
<i>Methanol plant wastewater</i>			
Sample 10	98.7	100.4	1.72
<i>Municipal wastewater treatment plant influent</i>			
Sample 11	399.3	403.1	0.95
<i>Municipal wastewater treatment plant effluent</i>			
Sample 12	47.2	48.7	3.18

4. Conclusion

In this study, the effect of preparation parameters on the photoelectrocatalytic performance of TiO₂ nanotube array electrodes including the electrolyte, anodic potential, anodic time, solution pH, calcination temperature and time were examined. The TiO₂ nanotube array electrode with highly photoelectrocatalytic properties was chosen as the working electrode. A photoelectrocatalytic COD determination method was studied by combining the optimized TiO₂ nanotube array electrodes, thin-cell reactor and photoelectrocatalytic techniques. It requires about 1–5 min to complete the oxidation of organics without further titration, much faster than the standard K₂Cr₂O₇ method (2–4 h). It consumes very limited intoxic and inexpensive supporting electrolyte, free from toxic chemicals which may cause secondary pollution. A wide dynamic working range from 0 to 850 mg/L can be achieved. At the same time, it shows good accuracy, stability and reproducibility.

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